

levied on goods like wheat, rice, fertilizers, cement, sugar, cloth etc. Specific duties are quite easy to administer, as they do not involve the evaluation of the goods. Imposition of specific duties enables the government to keep out of the complexities of prices.

However, specific duties cannot be levied on high-valued goods such as diamonds, jewellery, watches, T.V. sets, motor cars, works of art like paintings etc. These articles can be taxed either based on weight, the surface area covered or the number of articles.

b) Ad Valorem Tariff:

‘Ad Valorem’ is the Latin word that means ‘on the value.’

When the duty is levied as a fixed percentage of the value of the traded commodity, it is called an Ad Valorem tariff. Such duties are levied on the products the value of which is disproportionately higher compared to their physical characteristics such as weight or measurement. These duties are more equitable as the costly goods, generally consumed by the rich, bear a greater burden of duty, while the cheaper goods bought by the poor, bear a lesser burden of tariff.

c) Compound Tariff:

The compound tariff is a combination of specific and ad valorem tariffs. The compound tariff includes specific duty on each unit of the commodity along with a percentage of ad valorem duty. Compound tariffs bring a greater elasticity to revenues. They also ensure more effective protection for domestic industries.

2) Classification based on the Purpose which a Tariff serves:

Based on the purpose that tariff serves, Tariffs can be of two types:

- a) Revenue Tariff and
- b) Protective Tariff.

a) Revenue Tariff:

Sometimes the intention of the government behind imposing tariffs is revenue generation. Especially in developing countries, the government sometimes rely on tariffs as a source of revenue.

However, many times imposing tariffs also doesn't generate revenue. Tariff imposition may also lead to a shift of consumer demand from imported goods to domestic products, thus reducing the tariff collection. Usually, the rate of revenue tariff is kept low so that the revenue generated from the tariff is not curtailed because of reduced imports.

b) Protective Tariff:

When the purpose behind the imposition of a tariff is the protection of domestic industries from foreign competition, the tariff is called a protective tariff. As against revenue tariff, the rate of protective tariff is kept higher to reduce the imports. When the tariff rate is higher the protective effect of the tariff is also greater. A perfect protective tariff will prohibit imports. However, a case against protective tariffs is that because of zero competition from foreign producers, domestic producers will become inefficient and they might not be able to face competition even in long run.

3) Classification based on Discrimination:

Tariff systems can be classified into three categories based on their application in different countries.

- a) Single-Column tariff
- b) Double-Column tariff
- c) Triple-Column tariff

a) Single-Column Tariff:

The Single-Column Tariff also known as the uni-linear tariff system provides uniform tariff rates to all commodities irrespective of the country of origin. It is a simple tariff system and is easy to administer. It is a non-discriminatory tariff which sometimes becomes inelastic to adjust to the changing needs of the domestic industries.

b) Double-Column tariff:

Different tariff rates are charged for different countries in the Double-Column tariff system. The double-column tariff system is divided into two different types of tariffs: (a) General and conventional tariffs. The legislature fixes the general schedule at the very start while the conventional schedule is based on commercial treaties with other countries. (b) There are two autonomously determined schedules of tariff-the maximum and the minimum.

The minimum schedule applies to those countries who have taken the MFN (most favoured nation) pledge or signed an agreement. The maximum schedule applies to all other countries.

c) Triple-column tariff:

The triple-column tariff system consists of three autonomously determined tariff schedules-the general, the intermediate and the preferential. The general and intermediate tariff rates are like the maximum and minimum rates mentioned above under the Double-

Column tariff system. Preferential tariffs were once imposed between nations and their colonies.

4) **Classification based on Products:**

Tariffs can be classified on the basis whether a product is imported or exported. On this basis, the tariffs can be of two types:

- a) Import duties and
- b) Exports duties.

a) **Import Duties:**

If the home country imposes a tariff upon the products of foreign countries as they are imported, the tariff is known as an import tariff or import duty.

b) **Export Duties:**

Export duties are imposed on the commodities exported to be sold in foreign market, it is called an export tariff or export duty.

6.2.2 Effective Rate of Protection

Tariffs can be imposed on both final goods and intermediate goods as well. Tariffs on raw materials and intermediates impact the cost of production for domestic industry. Thus, we must consider tariff on raw material while assessing the extent of protection provided to the domestic industry in prevailing tariff structure. The extent of protection granted to the domestic industry by the given structure of nominal tariff rates is measured by the Effective Rate of protection (ERP). Under free trade, value added in the domestic industry would be determined by world market prices of the final good and imported inputs. And thus, effective rate of protection will be same as tariff rate. Also, if production of the final good does not need any intermediate goods, then the ERP equals the nominal rate of tariff on the final good. When a tariff is imposed, domestic prices can exceed the world market prices. Here, domestic value added and ERP would depend on the extent of price increase of the final good vis-à-vis that of intermediates along with the need of the intermediate goods in the production process.

ERP measures the change in value added in the domestic industry due to tariff protection. By definition,

$$ERP_F = (V_T - V_W) / V_W,$$

ERP_F is the ERP on the final good F; V_T is a value-added in the production of F after tariff; and V_W , is a value-added in the production of F at world market prices, i.e. V_W is the value added under 'free trade'. Remember, value added is the difference between the price of the 'final good produced by an industry, less the cost of inputs required per unit of production of the final good.

Let us consider a few examples to understand the concept of ERP clearly.

Consider a final good F whose world market price is Rs.10000. To produce one unit of F, you require one unit of an intermediate good I, whose world market price is Rs.5000. Thus V_w , in this case, is Rs.5000 (i.e. $10000-5000$).

Case 1 Suppose a tariff of 20% is imposed on the final good F, while there is no tariff on the intermediate, I. Assuming a small open economy, post tariff, the price of F in the domestic market rises by 20% (to Rs.12000) and V_T is Rs. 7000 (i.e., $12000-5000$). Hence ERP is 40% [i.e. $((7000-5000) / 5000) * 100$, which is much higher than the nominal rate of protection of 20%.

Case 2 Suppose there is no tariff on the final good F and a tariff of 10% is imposed on intermediate I. The small economy assumption means there will be no change in the world market price of I after the tariff, while its domestic price will increase by 10% (to Rs.5500). Now, V_T is Rs. 4500 (i.e., $10000-5500$) and ERP is - 10% [i.e. $((4500 -5000) / 5000) * 100$].

Following points can be kept in mind regarding ERP:

- if production of the final good does not need any intermediate goods, then the ERP will be equal to the nominal rate of tariff on the final good.
- if the tariff on final goods is higher than that on intermediate goods, then the ERP would be higher than the nominal rate of protection.
- ERP could be negative when tariffs on imports are too high compared to the tariff on the final good. When ERP is negative, it means that tariffs are reducing domestic value added as compared to free trade.

The measurement of ERP sometimes gets difficult because various inputs with different tariff rates are used to produce one commodity. Some other limitations of the concept of ERP is that it assumes fixed coefficients of production which do not allow for factor substitution possibilities. Also, the assumption of small open economy which cannot influence world prices is sometimes unrealistic.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Consider the following cases and describe whether ERP will be higher, lower or equal to the nominal rate of tariff:

Case I: If the production of the final good requires no intermediates:_____

Case II: If the tariff on final goods is higher than that on intermediates:_____

Case III: If the tariff on intermediate goods is higher than that on final goods.

- 2) Differentiate between Single column tariffs and double-column tariffs.

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- 3) List different categories of tariff barriers.

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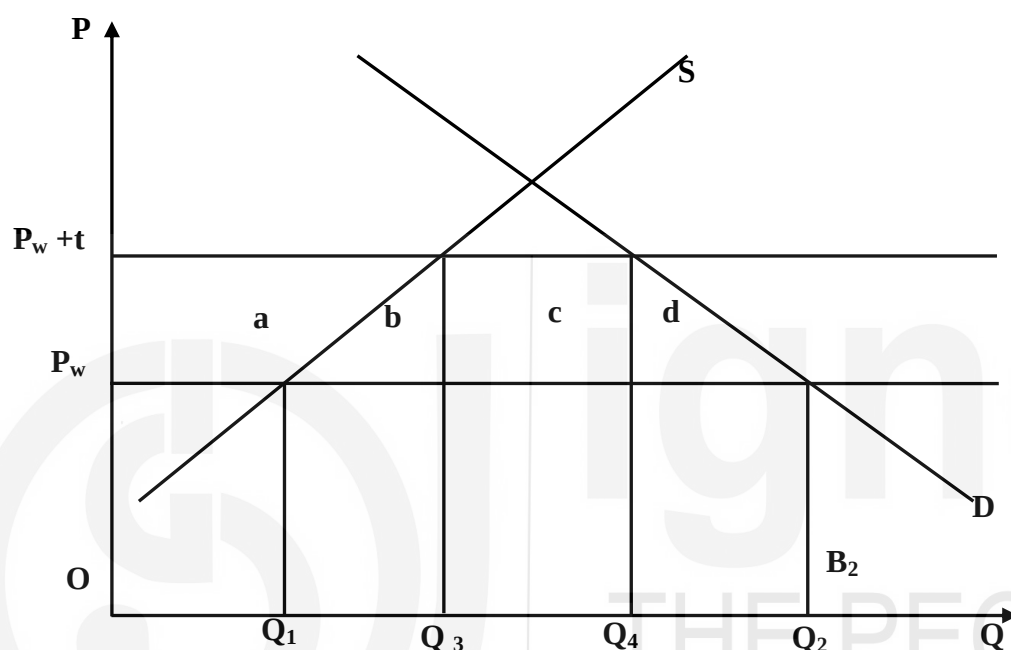
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6.2.3 Economic Impact of Import Tariff in partial equilibrium analysis

Using Figure 6.1, we will examine the impact of a tariff on various economic agents like consumers, producers and the government in a partial equilibrium framework. We assume that the country is a small open economy that cannot affect world market prices. This assumption implies that the tariff leaves the world market price of the good unaffected while raising its price in the domestic market. In Figure 6.1, we consider the domestic demand and supply curves of the imported good. The world market price of the good is P_w , which would prevail in the domestic market under free trade. Thus, with free trade, domestic production of the good would be Q_1 , domestic demand Q_2 , and the difference $Q_1 Q_2$ would be imported. A specific tariff rate of t per unit drives a wedge between the world price and the domestic market price of the imported good. Post-tariff prices in the domestic market rise to $(P_w + t)$. As a result, domestic production increases from Q_1 to Q_3 while consumption falls from Q_2 to Q_4 .

Now let us consider the costs imposed by the tariff. Owing to the tariff, domestic consumers suffer a loss in consumer surplus equal to the area $(a + b + c + d)$, compared to free trade. This loss arises because consumers must now pay a higher price $(P_w + t)$ for the good and also because they now consume Q_4Q_2 amount less of the good than with free trade. However, domestic producers gain from the price rise, with the area representing the increase in producer surplus. The government also gains from the tariff as it earns revenue. The imports are now Q_3Q_4 units, each paying a tariff of t per unit, so the tariff revenue is given by the rectangle with area c . The loss to consumers exceeds the gain to producers and the government taken together, indicating that the tariff results in a net social loss to society equal to the sum of areas b and d . In Figure 6.1, Area b measures the loss in social welfare from the production distortion due to the tariff. It represents the higher cost of producing Q_1Q_3 domestically rather than importing this amount; Area d is the welfare loss due to the consumption distortion created by the tariff. It represents the cost of not consuming Q_4Q_2 , an amount whose value to consumers exceeds the cost of importing it.

You should note that our partial equilibrium analysis of the impact of a tariff focuses simply on the market for the imported good. Repercussions on several important issues like terms of trade, BOP, factor markets etc., are left out of this framework, as in the case of our analysis of the benefits of free trade. In particular, our analysis does not attach any weight to employment in the sector being granted tariff protection. If the import-competing sector accounts for a significant share of total employment, the measures of welfare loss would have to be adjusted accordingly. In reality, employment is often the most important reason underlying the imposition of protectionist trade policies.



Source: Figure 6.3 in Sikdar (2003)

Figure 6.1: Economic impact of Import tariff in a partial Equilibrium analysis

6.2.4 General Equilibrium Analysis: Impact of Tariff on Volume of Trade and Terms of Trade (TOT)

When a country imposes a tariff, not only a specific product or sector but every sector of the economy gets affected in one way or the other. Import tariff changes the production and the consumption of other goods which in turn affect the composition of production and the allocation of resources across different sectors. Eventually, the volume of trade changes correspondingly. Further, a change in the Volume of trade may change the country's TOT (depending upon the size of the country and how the government spends the tax revenue). Let us examine the impact of the tariff on the Volume of trade and Terms of trade (TOT) in a small country and a large country.

1) General Equilibrium Analysis of Tariff in a Small Country:

When the tariff-imposing country is small, the domestic price of the importable commodity will rise by the full amount of tariff for the individual

consumers and producers in that small tariff-imposing country. The international price of the commodity will, however, remain unaffected. Let us assume that a small economy produces two commodities: Machinery (being imported) and textiles (being exported). When an import tariff is imposed, it will increase the price of machinery, which will encourage domestic production. But, if all the resources are fully employed, expansion in the production of machinery will require contraction in the production of textiles. Thus an import tariff will expand the machinery sector and contracts the export sector. On the other hand, the higher domestic price of machinery will shift consumption from machinery to textiles. Consequently, both the volume of exports and the volume of imports will decline. Thus the Volume of trade will fall.

Import tariff leads to welfare losses for a small country on two fronts: 1. Import tariff reallocates the resources from exporting the commodity to importing the commodity in which the country has a comparative disadvantage. 2. It alters the consumption pattern of the consumers as they have to pay a higher price for importing goods.

The general equilibrium analysis of tariffs in the case of a small country can be understood with the help of Fig. 6.2.

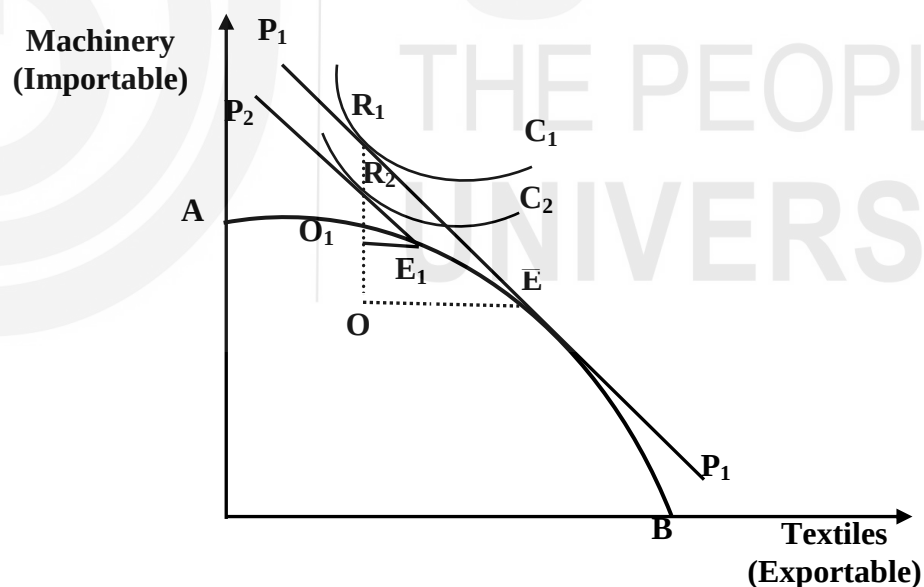


Figure 6.2: General Equilibrium Analysis of Tariff in a Small Country

In Fig. 6.2., the production possibility curve related to two commodities Textile and Machinery is AB. Under the assumptions of CRS technology and diminishing marginal productivities, PPF is drawn strictly concave downwards.

Initially, in the case of free international trade, P_1P_1 is the international exchange ratio line and the production equilibrium point is E.

The consumption equilibrium point is R_1 which lies on the community indifference curve C_1 . In this situation, country A exports OE quantity of textiles

and imports OR_1 quantity of machinery. If a tariff is imposed but the world prices of commodities remain the same, the international exchange ratio line is P_2E_1 , parallel to the original international exchange ratio line P_1P_1 .

Now production equilibrium shifts to E_1 where country A produces a large quantity of domestic machinery (importable goods). This is the production or protective effect of tariffs. The consumption equilibrium shifts from R_1 to R_2 where the international exchange ratio line P_2E_1 becomes tangent to a lower community indifference curve C_2 .

It shows that tariff has caused a reduction in the welfare of the tariff-imposing small country. The shift in consumption point from R_1 to R_2 signifies the consumption effect of the tariff. After the tariff, the country exports the O_1E_1 quantity of textiles and imports the R_2O_1 quantity of machinery. Both exports and imports are smaller than the free trade levels. Thus, tariffs for a small country is welfare reducing.

2) General Equilibrium Analysis of Tariff in a Large Country:

For a large tariff-imposing country, a tariff changes its TOT along with VOT. If the tariff-imposing country is large, reduced demand for imports due to tariff imposition may reduce the world demand for the concerned commodity to a great extent. Since the extent of reduced demand is high, it may reduce the international price of the concerned commodity. Thus, the price of an exporting commodity remains unchanged while that of importing commodity falls. This will cause a change in the international price ratio and will bring an improvement in the terms of trade of the tariff-imposing large country.

The production effect, consumption effect and terms of trade effect due to tariff can be explained through Fig. 6.3. Continuing with the example in Figure 6.2, a country produces two commodities: Machinery (being imported) and textiles (being exported)

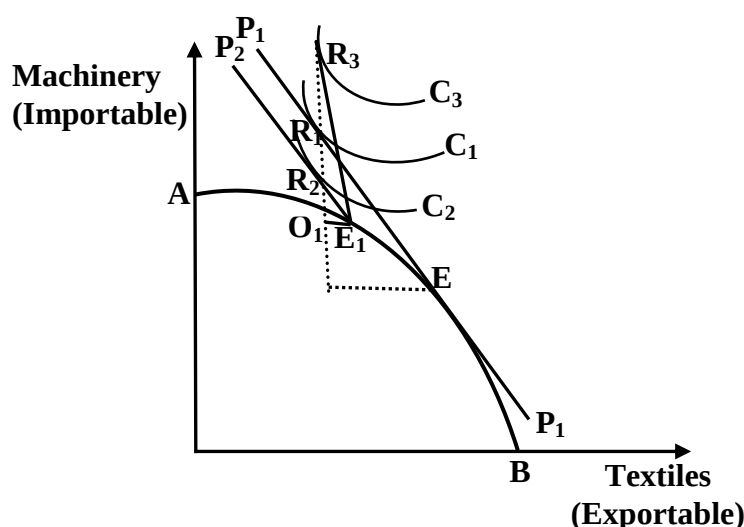


Figure 6.3: General Equilibrium Analysis of Tariff in a Large Country

When a large country imposes a tariff, the World price of the importable commodity falls (while the price of the exportable commodity is the same), shifting the international exchange price line to P_3E_1 . As you can observe in Figure 6.3, price line P_3E_1 is steeper than the exchange ratio line P_1P_1 or P_2E_1 . It simply means that the country can now buy its imports cheaper and sell its exports at the same price.

In this case, the production equilibrium occurs at E_1 and consumption equilibrium occurs at R_3 where P_3E_1 becomes tangent to the higher community indifference curve C_3 . The large tariff-imposing country imports R_3O_1 quantity of machinery and exports O_1E_1 quantity of textiles. A higher ratio of imports to exports indicates that the terms of trade have become favourable for the tariff imposing-country A.

The large country experiences production or protective effect as the domestic production of machinery increases owing to a shift in the production equilibrium from E to E_1 . Although, increased domestic production required the reallocation of resources the country has gained because of positive consumption and terms of trade effects.

The large country also experiences a welfare effect despite a reduction in specialisation and reallocation of resources towards import-substitute production. The welfare effect is reflected in the shift of consumption equilibrium to the highest community indifference curve C_3 . This is the positive consumption effect. Thus, large country imposing tariff benefits from the favourable TOT, despite the reduction in the volume of international trade.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Describe the economic impact of Import tariff in a partial Equilibrium analysis.

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- 2) What is the Impact of tariff on the Volume of trade and Terms of trade (TOT) in the case of a small country?

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3) Does a large country gain or lose from imposing a tariff? Justify.

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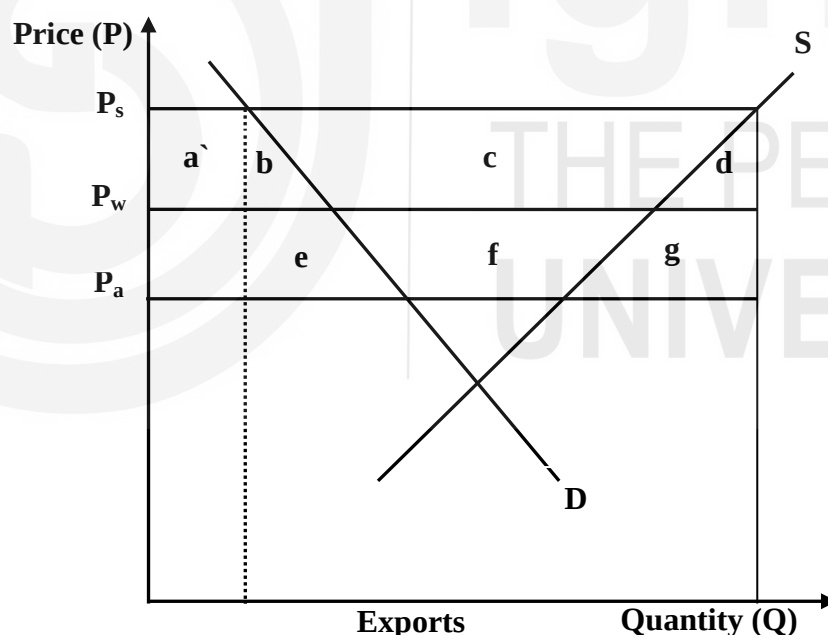
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6.3 EXPORT SUBSIDIES

Export subsidies are government policies that are implemented to encourage the export of goods by incentivizing local producers through easy and cheaper loans, direct payments, tax benefits etc. Like a tariff, an export subsidy can be either specific i.e. a fixed sum per unit or ad valorem i.e. the proportion of value exported. When the government provides an export subsidy, producers export up to a point where the domestic price is greater than the foreign price by the amount of subsidy. While export subsidies indeed help increase exports, there are costs associated with them. The effects of export subsidies on producers, consumers and the economy are explained in Figure 6.4



Source: Figure 6.4 in Sikdar (2003)

Figure 6.4: Effects of an export subsidy

When a country provides subsidies to exporters, the price will rise from P_w to P_s . At the same time, the price in importing countries falls from P_s to P_a , thus the price rise is less than the subsidy. In the exporting country, consumer loss is the area $a+b$ in Figure 6.4 as they have to pay higher. Producers' gain is reflected in the area $a+b+c$; the government subsidy is the area $b+c+d+e+f+g$. The net welfare loss is, therefore the sum of the area $b+d+e+f+g$. Also, unlike in the case of tariff, export subsidies worsens the terms of trade by lowering the International